Contact-Preserving Barrier Compression and Composition-Refined Enumeration of Rational Dyck Paths

Alp Eren Bütün
Independent Researcher

August 2026

Abstract

We develop a contact-preserving barrier-compression method for refined enumeration of monotone sequences and rational Dyck paths. For a nondecreasing upper barrier, deleting an arbitrary finite set of forbidden values induces an order-preserving compression of both the ambient alphabet and the barrier. A multivariate Lindström–Gessel–Viennot determinant then records the exact positions at which the original sequence meets its barrier. Inclusion–exclusion adds arbitrary required values and simultaneous first-missing constraints.

The method arose from a concrete arithmetic route. A completion-aware deterministic pushdown automaton for the binary balance relation $N_1 = 2N_0$ suggested the scalar residual $\rho_{p,q} = qN_1 - pN_0$. Under $m = gq$ and $n = gp$, the rational-path rank satisfies $R = g\rho_{p,q}$, so the low-rank return condition becomes an upper-barrier contact in coarea coordinates. The pushdown construction is retained explicitly as the discovery origin, while all enumeration proofs are self-contained combinatorics.

For rational Dyck paths with a fixed ordered vertical-run composition α, we obtain: an explicit single determinant for prescribed ordinary run and return values; finite signed determinant sums for prescribed ratio-run together with exact return components; simultaneous ordinary run, ratio-run, and exact return-set enumeration; and a single determinant for the common cardinality of every class on a fixed ratio-signature skew diagonal. The latter yields a positive complete-homogeneous expansion of the fixed-signature polynomial. We also apply the framework to paths below arbitrary northeast boundaries, partial Dyck paths, and column-increasing rational parking-function labelings. Exhaustive computation independently verifies the general barrier theorem and all four rational-path specializations over broad finite ranges, with no mismatches.

Keywords. rational Dyck paths; deterministic pushdown automata; weighted residual; barrier compression; run statistic; ratio-run; returns; exact contacts; Lindström–Gessel–Viennot lemma; binomial determinant; inclusion–exclusion; parking functions.

1 Introduction

Rational Dyck paths are lattice paths from $(0, 0)$ to (m, n), with north and east steps, that remain weakly above the line $y = nx/m$. They interpolate between classical Dyck paths, Fuss–Catalan paths, and the broader rational Catalan setting; see, for example, Armstrong, Loehr, and Warrington [1]. Dai, Fu, and Qiu introduced rational run and ratio-run statistics and proved that each has a symmetric joint distribution with the return statistic, by composition-preserving involutions [2]. Their concluding remarks explicitly ask for formulas enumerating rational Dyck paths with prescribed run values and returns.

The present paper develops a reduction framework that makes substantially finer enumerations explicit. Its central mechanism is *contact-preserving barrier compression*. A lattice path, or more generally a weakly increasing integer sequence, is encoded beneath a nondecreasing upper barrier. Prescribed forbidden values are deleted from the ordered alphabet, and the barrier is compressed accordingly. The compression is order preserving and, crucially, retains the information needed to detect the rows at which the original sequence meets its upper barrier. Classical Lindström–Gessel–Viennot and Kreweras-type determinants may then enumerate the compressed objects while marking exact boundary contacts [4, 3, 5].

The determinant technology itself is classical. Likewise, order-preserving relabeling and inclusion–exclusion are classical ingredients. The contribution here is their use as one contact-sensitive reduction: arbitrary presence and absence conditions, exact contact sets, and simultaneous first-missing statistics can be converted into explicit determinant enumeration without losing the boundary-contact statistic.

The method is motivated by, but logically independent of, the author's preceding pushdown and run–return work. The discovery route began with a completion-aware deterministic pushdown automaton for the weighted binary balance relation $N_1 = 2N_0$. The arithmetic invariant extracted from that machine led to the general residual

$$\rho_{p,q} = qN_1 - pN_0.$$

For a rational Dyck rectangle with

$$m = gq, \qquad n = gp,$$

this residual is precisely the usual rational rank up to the factor g:

$$R(x, y) = my - nx = g(qy - px) = g\rho_{p,q}.$$

Consequently the low-rank return condition becomes a low-residual condition, and in coarea coordinates this becomes an upper-barrier contact. At the unrestricted level this observation yields explicit run–return enumeration; the present paper extracts the underlying mechanism and pushes it to composition, exact-contact, simultaneous-statistic, signature, arbitrary-boundary, and labeled settings.

For a fixed ordered vertical-run composition $\alpha = (\alpha_1, \ldots, \alpha_d) \vDash n$, the paper proves four core enumeration results. First, the ordinary run–return classes reduce to a single bounded-sequence determinant. Second, ratio-run is encoded as a sampled first-missing condition and enumerated jointly with the exact set of return components. Third, ordinary run and ratio-run are prescribed simultaneously, again together with exact returns. Fourth, at fixed ratio-signature, the Dai–Fu–Qiu fixed-signature bijections reduce the class to a minimal-return representative; there the signature becomes a forced prefix and the free suffix is counted by a single determinant. Two applications then show that the method is not tied to a rational slope and that the fixed-composition formulas transfer directly to column-increasing parking-function labelings.

The paper is organized so that the automata-theoretic origin remains visible but is not used as a black box. Section 2 records the 2:1 DPDA and the residual-to-rank bridge. Section 3 proves the general contact-preserving compression theory. Section 4 develops fixed-composition coordinates and the effective barrier needed after deleting strict component columns. Sections 5–8 contain the four core enumerations. Sections 9 and 10 give the applications, Section 11 reports exhaustive verification, and Section 12 concludes.

2 Discovery origin: from pushdown residuals to barrier compression

The results after this section are purely combinatorial, but the route by which the relevant coordinates were found is useful for explaining why the barrier-contact formulation arises naturally.

2.1 The motivating completion-aware 2:1 DPDA

Consider

$$L_{2,1} = \{w \in \{0,1\}^* : N_1(w) = 2N_0(w)\}.$$

The completion-aware deterministic pushdown automaton in Figure 1 is the concrete starting point of the present line of work. The center/left/right core stores unresolved completion demand jointly in finite control and in the stack. A transition label $a, A/\gamma$ means that input a is consumed and the top stack symbol A is replaced by γ; the leftmost work symbol is the stack top. The symbol Z_0 is the bottom marker. The dashed boxes contain only right-endmarker completion operations and are not ordinary ε-moves during binary-input processing.

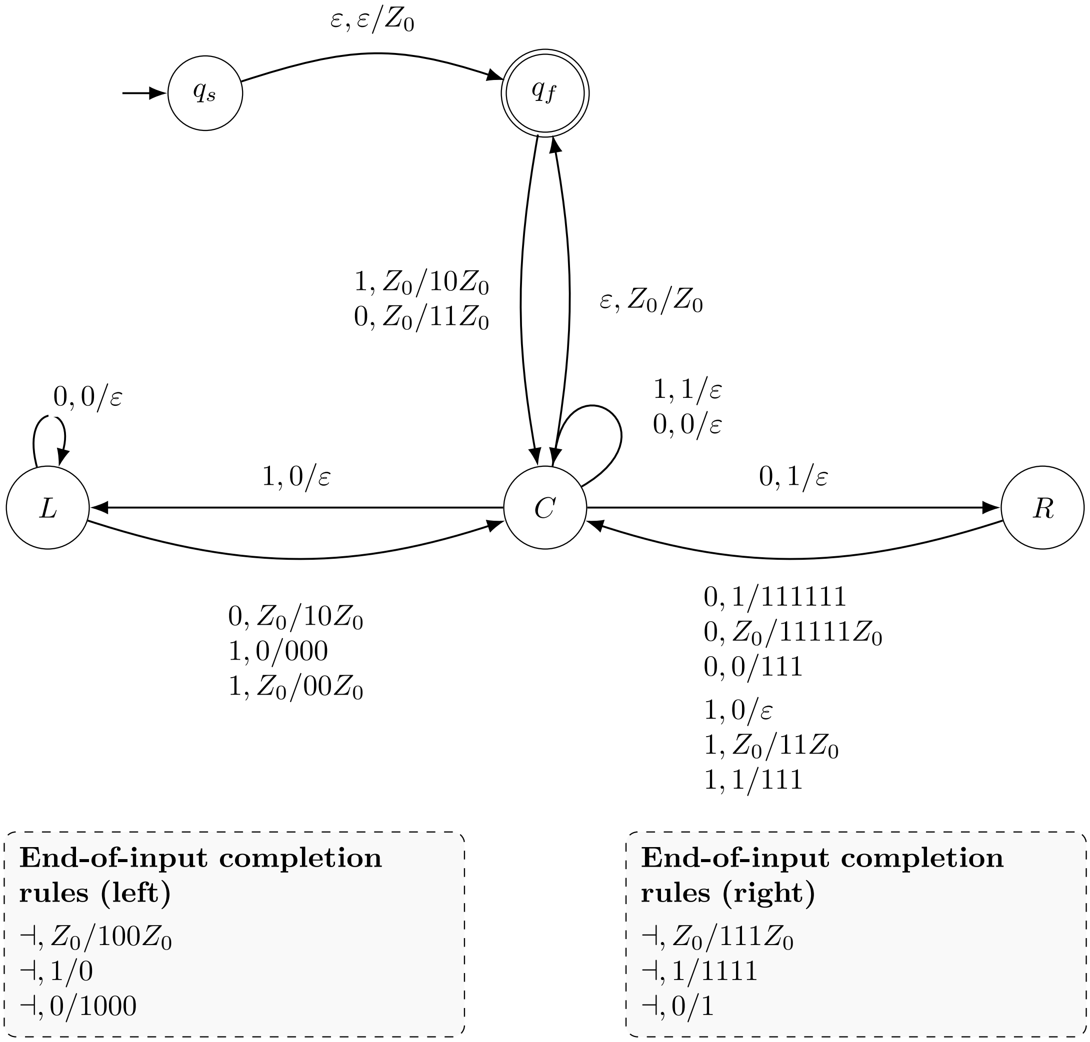

Figure 1: The completion-aware 2:1 deterministic pushdown automaton that motivated the scalar residual viewpoint. Solid arcs process binary input; dashed boxes contain only right-endmarker completion rules. The machine is included to record the concrete discovery origin and is not a hypothesis of the determinant theory.

Ignoring the bottom marker, let $S \in \{0,1\}^*$ denote the logical stack. Define

$$D(S) = 2N_0(S) - N_1(S), \qquad \Phi(C) = 0, \quad \Phi(L) = 3, \quad \Phi(R) = -3.$$

The processed input prefix has signed residual

$$\rho_{2,1}(w) = N_1(w) - 2N_0(w),$$

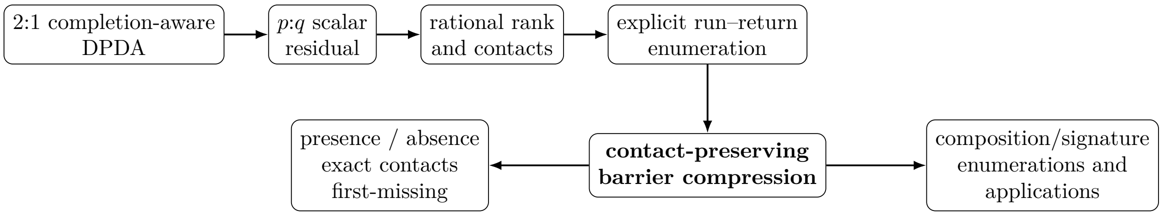

Figure 2: Discovery and abstraction route. The pushdown and residual steps explain the origin of the contact viewpoint; the theory of this paper begins with the abstraction to contact-preserving barrier compression.

and a direct check of the displayed binary transitions gives the distributed invariant

$$\rho_{2,1}(w) = D(S) + \Phi(q_{\text{state}}). \tag{1}$$

Thus the unresolved arithmetic is not tied to one particular stack word: part of the residual may be carried by finite control. This representation-independent scalar quantity is what survives the passage to the general balance relation.

2.2 The general p:q residual and rational rank

For coprime positive integers p, q, write

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\}.$$

The canonical scalar imbalance is

$$\boxed{\rho_{p,q}(w) = qN_1(w) - pN_0(w).} \tag{2}$$

Reading 1 increases $\rho_{p,q}$ by q, while reading 0 decreases it by p. For the present paper, the machine-specific completion semantics are not needed beyond the scalar residual extracted from the 2:1 construction.

Encode an east step of a lattice path by 0 and a north step by 1. If a prefix ends at (x, y), then $x = N_0$ and $y = N_1$. Write

$$g = \gcd(m, n), \qquad m = gq, \qquad n = gp.$$

Then

$$\boxed{R(x, y) = my - nx = g(qy - px) = g\rho_{p,q}.} \tag{3}$$

Because a rational Dyck path has $R \geq 0$, the return condition $R < n = gp$ is equivalent to

$$\boxed{0 \leq \rho_{p,q} < p.} \tag{4}$$

In coarea coordinates this becomes an upper-barrier contact, as proved directly later.

Remark 2.1 (Role of the pushdown model)**.** The pushdown construction records the concrete discovery origin of the residual and contact viewpoints. It is not a logical prerequisite for the determinant theory developed below. Starting with Section 3, all results are proved in terms of monotone sequences, barrier compression, nonintersecting-path enumeration, and inclusion–exclusion.

3 Contact-preserving barrier compression

Throughout this section let

$$\mathbf{b} = (b_1, \dots, b_n), \qquad 0 \le b_1 \le \cdots \le b_n,$$

and define

$$\mathcal{A}_{\mathbf{b}} = \{u \in \mathbb{Z}^n_{\ge 0} : 0 \le u_1 \le \cdots \le u_n,\ u_i \le b_i\}.$$

For $u \in \mathcal{A}_{\mathbf{b}}$, set

$$\mathrm{Val}(u) = \{u_1, \dots, u_n\}, \qquad C_{\mathbf{b}}(u) = \{i \in [n] : u_i = b_i\}.$$

Presence and absence concern values of the sequence; contacts concern row positions. Keeping these two kinds of information separate is the central point of the method. We use the convention $\binom{a}{b} = 0$ whenever $b < 0$ or $b > a$.

3.1 Deleting forbidden values

Let $F \subset \mathbb{Z}_{>0}$ be finite and define

$$\phi_F(t) = t - \#(F \cap [1, t]), \qquad c_i(F) = \phi_F(b_i), \qquad \epsilon_i(F) = \mathbf{1}_{\{b_i \notin F\}}. \tag{5}$$

Lemma 3.1 (Contact-preserving value compression)**.** *The coordinatewise map*

$$\Phi_F(u_1, \dots, u_n) = (\phi_F(u_1), \dots, \phi_F(u_n))$$

is a bijection

$$\{u \in \mathcal{A}_{\mathbf{b}} : \mathrm{Val}(u) \cap F = \varnothing\} \longrightarrow \mathcal{A}_{\mathbf{c}(F)}.$$

Moreover, if $b_i \notin F$, then

$$u_i = b_i \iff \phi_F(u_i) = c_i(F),$$

whereas if $b_i \in F$ then no F-avoiding sequence can contact the original barrier in row i.

Proof. The restriction of ϕ_F to $\mathbb{Z}_{\ge 0} \setminus F$ is the unique order-preserving bijection from the allowed nonnegative integers onto $\mathbb{Z}_{\ge 0}$. Hence an F-avoiding weakly increasing sequence remains weakly increasing after applying ϕ_F, and $u_i \le b_i$ implies $\phi_F(u_i) \le \phi_F(b_i) = c_i(F)$. Conversely, applying the inverse order isomorphism coordinatewise to a sequence below $\mathbf{c}(F)$ gives an F-avoiding weakly increasing sequence below $\mathbf{b}$. If $b_i \notin F$, injectivity on the allowed alphabet gives $\phi_F(u_i) = \phi_F(b_i)$ if and only if $u_i = b_i$. If $b_i \in F$, that original contact is forbidden. $\square$

3.2 Multivariate contacts

Define

$$D_{\mathbf{b},F}(z_1, \dots, z_n) = \det_{1 \le i,j \le n} \left[\binom{c_i(F) + 1}{j - i + 1} + (z_i - 1)\epsilon_i(F) \binom{c_i(F)}{j - i} \right]. \tag{6}$$

Theorem 3.2 (Multivariate contact determinant)**.** *For every finite $F \subset \mathbb{Z}_{>0}$,*

$$\boxed{D_{\mathbf{b},F}(\mathbf{z}) = \sum_{\substack{u \in \mathcal{A}_{\mathbf{b}} \\ \mathrm{Val}(u) \cap F = \varnothing}} \prod_{i \in C_{\mathbf{b}}(u)} z_i.} \tag{7}$$

Proof. Write $c_i = c_i(F)$ and $\epsilon_i = \epsilon_i(F)$. Consider the directed acyclic lattice network with east steps $(1, 0)$ and northeast steps $(1, 1)$, sources

$$S_i = (-c_i - 1, i),$$

and targets

$$T_j = (0, j+1).$$

A path from S_i to T_j has $c_i + 1$ horizontal steps and $j - i + 1$ northeast steps, so there are $\binom{c_i+1}{j-i+1}$ such paths. If $\epsilon_i = 1$, give weight z_i to the northeast edge leaving S_i and weight 1 to all other edges; if $\epsilon_i = 0$, give all initial edges weight 1. The number of S_i–T_j paths using the initial northeast edge is $\binom{c_i}{j-i}$, giving exactly the matrix entry in (6).

By the Lindström–Gessel–Viennot lemma [3], the determinant is the signed generating function of vertex-disjoint path families. Since c_i is nondecreasing, the source–target configuration is nonpermutable: an inversion of targets would reverse vertical order over a common horizontal interval and force an intersection. Thus only the identity matching contributes.

An identity path $\pi_i : S_i \to T_i$ has one northeast step and c_i east steps. Let e_i be the number of east steps before that northeast step and set $v_i = c_i - e_i$. Adjacent paths are vertex-disjoint exactly when $v_i \le v_{i+1}$. Hence nonintersecting identity families are in bijection with

$$0 \le v_1 \le \cdots \le v_n, \qquad v_i \le c_i.$$

Moreover, π_i uses its initial northeast edge exactly when $v_i = c_i$. Lemma 3.1 identifies the weighted instances of these compressed contacts with the original contacts $u_i = b_i$. Therefore the family corresponding to u has weight $\prod_{i \in C_{\mathbf{b}}(u)} z_i$. □

For exact contacts define

$$B_{ij}^F = \epsilon_i(F) \binom{c_i(F)}{j-i}, \qquad A_{ij}^F = \binom{c_i(F)+1}{j-i+1} - B_{ij}^F. \tag{8}$$

For $J \subseteq [n]$, let $\mathsf{M}_J^{\mathbf{b},F}$ be the matrix whose ith row is B_i^F for $i \in J$ and A_i^F for $i \notin J$.

Corollary 3.3 (Exact contact-set determinant)**.** *For every $J \subseteq [n]$,*

$$\boxed{\#\{u \in \mathcal{A}_{\mathbf{b}} : \mathrm{Val}(u) \cap F = \varnothing,\ C_{\mathbf{b}}(u) = J\} = \det \mathsf{M}_J^{\mathbf{b},F}.}$$

Proof. The ith row in (6) is $A_i^F + z_i B_i^F$. Expand by row multilinearity and compare the coefficient of $\prod_{i \in J} z_i$ with Theorem 3.2. □

3.3 Presence, absence, and simultaneous first-missing statistics

Let $R, F \subset \mathbb{Z}_{>0}$ be finite and disjoint. Set

$$G_{\mathbf{b};R,F}(\mathbf{z}) = \sum_{T \subseteq R} (-1)^{|T|} D_{\mathbf{b},F \cup T}(\mathbf{z}). \tag{9}$$

Theorem 3.4 (Presence–absence contact calculus)**.** *For finite disjoint $R, F \subset \mathbb{Z}_{>0}$,*

$$\boxed{G_{\mathbf{b};R,F}(\mathbf{z}) = \sum_{\substack{u \in \mathcal{A}_{\mathbf{b}} \\ R \subseteq \mathrm{Val}(u),\ \mathrm{Val}(u) \cap F = \varnothing}} \ \prod_{i \in C_{\mathbf{b}}(u)} z_i.}$$

Consequently, for every $J \subseteq [n]$,

$$\boxed{\#\{u \in \mathcal{A}_{\mathbf{b}} : R \subseteq \mathrm{Val}(u),\ \mathrm{Val}(u) \cap F = \varnothing,\ C_{\mathbf{b}}(u) = J\} = \sum_{T \subseteq R} (-1)^{|T|} \det \mathsf{M}_J^{\mathbf{b},F \cup T}.} \tag{10}$$

Proof. Among the objects avoiding F, require every $r \in R$ to occur. Inclusion–exclusion over the events $r \notin \mathrm{Val}(u)$ gives (9); Theorem 3.2 supplies the weighted determinant, and Corollary 3.3 gives (10). □

Let $s^{(1)}, \dots, s^{(q)}$ be injective sequences of positive integers and define

$$\mathrm{fm}_{s^{(h)}}(u) = \min\{a \ge 1 : s_a^{(h)} \notin \mathrm{Val}(u)\}.$$

For prescribed $a_1, \dots, a_q$, set

$$R(\mathbf{a}) = \bigcup_{h=1}^{q} \{s_1^{(h)}, \dots, s_{a_h-1}^{(h)}\}, \qquad F(\mathbf{a}) = \{s_{a_1}^{(1)}, \dots, s_{a_q}^{(q)}\}.$$

Corollary 3.5 (Simultaneous first-missing statistics)**.** *If $R(\mathbf{a}) \cap F(\mathbf{a}) \neq \varnothing$, no sequence satisfies all prescribed first-missing conditions. Otherwise, for every $J \subseteq [n]$,*

$$\boxed{\#\{u \in \mathcal{A}_{\mathbf{b}} : \mathrm{fm}_{s^{(h)}}(u) = a_h \ \forall h,\ C_{\mathbf{b}}(u) = J\} = \sum_{T \subseteq R(\mathbf{a})} (-1)^{|T|} \det \mathsf{M}_J^{\mathbf{b}, F(\mathbf{a}) \cup T}.}$$

Proof. Each condition $\mathrm{fm}_{s^{(h)}}(u) = a_h$ requires $s_1^{(h)}, \dots, s_{a_h-1}^{(h)}$ to be present and $s_{a_h}^{(h)}$ to be absent. Combine these conditions and apply Theorem 3.4. □

Remark 3.6. The ordinary probe $s_a = a$ and the sampled probe $s_a = \delta a$ are immediate special cases. No rational-slope, Dyck-path, or automata assumption occurs anywhere in this section.

4 Fixed-composition rational Dyck paths

Assume henceforth that $m > n \ge 1$ and let

$$\alpha = (\alpha_1, \dots, \alpha_d) \vDash n, \qquad A_0 = 0, \quad A_i = \alpha_1 + \cdots + \alpha_i.$$

Let $\mathcal{D}_{m,\alpha}$ be the set of $m \times n$ rational Dyck paths with ordered vertical-run composition α.

4.1 Component-column coordinates and returns

For $P \in \mathcal{D}_{m,\alpha}$, let x_i be the horizontal coordinate of the initial vertex of the ith vertical component and set

$$\beta_i = \left\lfloor \frac{mA_{i-1}}{n} \right\rfloor. \tag{11}$$

Lemma 4.1 (Fixed-composition component encoding)**.** *The map $P \mapsto (x_1, \dots, x_d)$ is a bijection from $\mathcal{D}_{m,\alpha}$ onto*

$$\mathcal{X}_{m,\alpha} = \{(x_1, \dots, x_d) : 0 = x_1 < x_2 < \cdots < x_d,\ x_i \le \beta_i\}.$$

Proof. The ith component starts at (x_i, A_{i-1}). The Dyck condition gives $A_{i-1} \ge nx_i/m$, hence $x_i \le \beta_i$. Distinct vertical components are separated by east steps, so their columns are strictly increasing. Conversely, from such a column sequence reconstruct

$$N^{\alpha_1} E^{x_2 - x_1} N^{\alpha_2} \cdots E^{x_d - x_{d-1}} N^{\alpha_d} E^{m - x_d}.$$

Each horizontal segment stays above the diagonal because its right endpoint satisfies the corresponding bound $x_{i+1} \le \lfloor mA_i/n \rfloor$. □

Let $\mathrm{RetComp}(P)$ be the set of component indices whose first north step is a return.

Lemma 4.2 (Return components are barrier contacts)**.** *For $P \leftrightarrow (x_1, \dots, x_d)$,*

$$\boxed{i \in \mathrm{RetComp}(P) \iff x_i = \beta_i.}$$

Consequently $\mathrm{ret}(P) = |\,\mathrm{RetComp}(P)|$.

Proof. At the start of component i, the rank is

$$R_i = mA_{i-1} - nx_i.$$

Write $mA_{i-1} = n\beta_i + r_i$ with $0 \le r_i < n$. Then

$$R_i = r_i + n(\beta_i - x_i).$$

Since $x_i \le \beta_i$, one has $R_i < n$ exactly when $x_i = \beta_i$. This is precisely the return condition of Dai–Fu–Qiu [2]. A later north step in the same component has rank at least $R_i + m \ge n$, so a component contributes at most one return. □

Corollary 4.3 (Exact north-step return positions). *If $J = \mathrm{RetComp}(P)$, then*

$$\boxed{\mathrm{ReturnSet}(P) = \{A_{i-1} + 1 : i \in J\}.}$$

4.2 Forbidden columns and the effective barrier

For finite $F \subset \mathbb{Z}_{>0}$, set

$$c_i(F) = \beta_i - \#(F \cap [1, \beta_i]), \qquad g_i(F) = c_i(F) - (i-1),$$

and define the suffix-minimum effective barrier

$$\boxed{h_i(F) = \min_{j \ge i} g_j(F).} \tag{12}$$

Then $h_1(F) \le \cdots \le h_d(F)$.

Lemma 4.4 (Effective-barrier regularization). *Let*

$$\mathcal{X}_{m,\alpha}(F) = \{x \in \mathcal{X}_{m,\alpha} : \{x_1, \ldots, x_d\} \cap F = \varnothing\}.$$

If $h_1(F) < 0$, the class is empty. If $h_1(F) \ge 0$, the maps

$$y_i = \phi_F(x_i), \qquad v_i = y_i - (i-1)$$

give a bijection

$$\mathcal{X}_{m,\alpha}(F) \longleftrightarrow \{0 \le v_1 \le \cdots \le v_d : v_i \le h_i(F)\}.$$

Proof. Because ϕ_F is strictly increasing on the allowed alphabet, strict x-inequalities are equivalent to strict y-inequalities. After the shift $v_i = y_i - (i-1)$, strictness becomes weak monotonicity and the raw upper bound is $v_i \le g_i(F)$. For a weakly increasing sequence, the full system $v_j \le g_j(F)$ is equivalent to $v_i \le \min_{j \ge i} g_j(F) = h_i(F)$ for every i. Since $\beta_1 = 0$, feasibility forces $h_1(F) = 0$; if $h_1(F) < 0$ no nonnegative sequence exists. □

Define the surviving-contact indicator

$$\eta_i(F) = \mathbf{1}_{\{\beta_i \notin F,\ h_i(F) = g_i(F)\}}. \tag{13}$$

Lemma 4.5 (Surviving return contacts). *Under Lemma 4.4,*

$$\boxed{x_i = \beta_i \iff \eta_i(F) = 1 \text{ and } v_i = h_i(F).}$$

Proof. If $x_i = \beta_i$, then $\beta_i \notin F$ and $v_i = c_i(F) - (i-1) = g_i(F)$. Since $v_i \le h_i(F) \le g_i(F)$, equality is forced. Conversely, if $\eta_i(F) = 1$ and $v_i = h_i(F) = g_i(F)$, then $y_i = c_i(F) = \phi_F(\beta_i)$; injectivity on the allowed alphabet gives $x_i = \beta_i$. □

Define

$$D_{\alpha,F}(z_1,\dots,z_d)=\det_{1\le i,j\le d}\left[\binom{h_i(F)+1}{j-i+1}+(z_i-1)\eta_i(F)\binom{h_i(F)}{j-i}\right] \tag{14}$$

when $h_1(F)\ge 0$, and set it equal to zero otherwise. Also define

$$B^{\alpha,F}_{ij}=\eta_i(F)\binom{h_i(F)}{j-i},\qquad A^{\alpha,F}_{ij}=\binom{h_i(F)+1}{j-i+1}-B^{\alpha,F}_{ij},$$

and let $M^{\alpha,F}_J$ use row $B^{\alpha,F}_i$ when $i\in J$ and $A^{\alpha,F}_i$ otherwise.

Theorem 4.6 (Fixed-composition forbidden-column determinant)**.** *For finite $F\subset\mathbb{Z}_{>0}$,*

$$\boxed{D_{\alpha,F}(\mathbf{z})=\sum_{\substack{P\in\mathcal{D}_{m,\alpha}\\ \{x_i(P)\}\cap F=\varnothing}}\ \prod_{i\in\mathrm{RetComp}(P)} z_i.}$$

Consequently,

$$\boxed{\#\{P\in\mathcal{D}_{m,\alpha}:\{x_i(P)\}\cap F=\varnothing,\ \mathrm{RetComp}(P)=J\}=\det M^{\alpha,F}_J.} \tag{15}$$

Proof. Lemma 4.4 gives a weakly increasing sequence below $\mathbf{h}(F)$. Apply Theorem 3.2 to that effective barrier, but mark row i only when $\eta_i(F)=1$. Lemma 4.5 identifies exactly those marked effective contacts with the original return components. The exact-set formula follows by row multilinearity. □

Remark 4.7. For $F=\varnothing$, $g_i=\beta_i-(i-1)$ is already nondecreasing because $m>n$ and every part of α is positive. Thus $h_i=g_i$ and $\eta_i=1$. The suffix-minimum regularization is needed only after forbidden values are deleted.

5 Explicit fixed-composition run–return enumeration

Dai–Fu–Qiu define a statistic $\mathrm{rr}(\alpha,m)$ depending only on the composition and m; if $I_1(\alpha)$ is the length of the maximal initial string of parts equal to 1, then [2]

$$r_\alpha:=\mathrm{rr}(\alpha,m)=\min\left\{I_1(\alpha)+1,\left\lceil\frac{n}{m-n}\right\rceil\right\}. \tag{16}$$

Every path of composition α has $\mathrm{run}(P),\mathrm{ret}(P)\ge r_\alpha$. Their fixed-composition bijections imply that for $K=k+\ell-r_\alpha$,

$$|\mathcal{D}^{(k,\ell)}_{m,\alpha}|=|\mathcal{D}^{(K,r_\alpha)}_{m,\alpha}|, \tag{17}$$

whenever the class is nonempty [2, Theorem 3.2 and Corollary 3.5].

For $K<d$, put $p=d-K$ and define

$$B^{(\alpha,K)}_i=\beta_{K+i}-(K+i)-1=\left\lfloor\frac{mA_{K+i-1}}{n}\right\rfloor-(K+i)-1,\qquad 1\le i\le p. \tag{18}$$

Lemma 5.1 (Minimal-return suffix barrier)**.** *For a path in $\mathcal{D}^{(K,r_\alpha)}_{m,\alpha}$, the first K component columns are forced:*

$$x_i=i-1\qquad(1\le i\le K).$$

If $K<d$, then after the shift

$$v_i=x_{K+i}-(K+i),$$

the free suffix is in bijection with

$$0\le v_1\le\cdots\le v_p,\qquad v_i\le B^{(\alpha,K)}_i.$$

Moreover $B^{(\alpha,K)}_1\le\cdots\le B^{(\alpha,K)}_p$.

Proof. In the minimal-return representative, Dai–Fu–Qiu use the decomposition $M_1 \cdots M_K ET$ with $M_i = N^{\alpha_i}E$ [2, Corollary 3.5]. Hence the first K component columns are $0, 1, \ldots, K-1$ and the forced separator implies $x_{K+1} \geq K+1$. Since the suffix contributes no further returns, Lemma 4.2 gives $x_i \leq \beta_i - 1$ for $i > K$. The stated shift yields the weak sequence and its barrier. Finally

$$B_{i+1}^{(\alpha,K)} - B_i^{(\alpha,K)} = \beta_{K+i+1} - \beta_{K+i} - 1 \geq 0,$$

because $m > n$ and $\alpha_{K+i} \geq 1$. $\square$

Theorem 5.2 (Explicit composition-refined run–return enumeration)**.** *Let $k, \ell \geq 1$ and set $K = k + \ell - r_\alpha$. If any of*

$$k < r_\alpha, \qquad \ell < r_\alpha, \qquad K > d$$

holds, the class is empty. If $K < d$ and $B_1^{(\alpha,K)} < 0$, it is likewise empty. Otherwise,

$$\boxed{|\mathcal{D}_{m,\alpha}^{(k,\ell)}| = \det_{1 \leq i,j \leq d-K} \left[\binom{B_i^{(\alpha,K)} + 1}{j - i + 1} \right].} \tag{19}$$

For $K = d$, the empty determinant is interpreted as 1.

Proof. Use the Dai–Fu–Qiu reduction (17), then apply the classical bounded weak-sequence determinant to the suffix barrier in Lemma 5.1. $\square$

Set

$$C_{\alpha,K} = \det_{1 \leq i,j \leq d-K} \left[\binom{B_i^{(\alpha,K)} + 1}{j - i + 1} \right].$$

Corollary 5.3 (Positive complete-homogeneous expansion)**.** *With*

$$F_{m,\alpha}(u,v) = \sum_{P \in \mathcal{D}_{m,\alpha}} u^{\mathrm{run}(P)} v^{\mathrm{ret}(P)},$$

one has

$$\boxed{F_{m,\alpha}(u,v) = (uv)^{r_\alpha} \sum_{K=r_\alpha}^{d} C_{\alpha,K} h_{K-r_\alpha}(u,v),} \tag{20}$$

where $h_j(u,v) = u^j + u^{j-1}v + \cdots + v^j$.

Proof. For fixed K, the feasible statistic pairs are $(r_\alpha + a, K - a)$ for $0 \leq a \leq K - r_\alpha$, and Theorem 5.2 gives the common cardinality $C_{\alpha,K}$. Summing their monomials gives $(uv)^{r_\alpha} h_{K-r_\alpha}(u,v)$. $\square$

Example 5.4. Let $(m,n) = (11,5)$ and $\alpha = (2,1,1,1)$. Then $\beta = (0,4,6,8)$ and $r_\alpha = 1$. For $(\mathrm{run}, \mathrm{ret}) = (1,2)$, $K = 2$ and the suffix barrier is $(2,3)$. Hence

$$|\mathcal{D}_{11,(2,1,1,1)}^{(1,2)}| = \det \begin{pmatrix} 3 & 3 \\ 1 & 4 \end{pmatrix} = 9.$$

Remark 5.5. The determinant in Theorem 5.2 is classical. The new enumerative step is the reduction from a fixed ordered composition and prescribed $(\mathrm{run}, \mathrm{ret})$ to a single bounded monotone suffix. In particular, it makes the cardinality in the partial-path characterization of Dai–Fu–Qiu Corollary 3.5 explicit.

6 Composition-refined ratio-run and returns

Set

$$\kappa = \left\lfloor \frac{m}{n} \right\rfloor .$$

For $P \in \mathcal{D}_{m,\alpha}$, let $X(P) = \{x_1(P), \dots, x_d(P)\}$.

Lemma 6.1 (Coarea values and component columns)**.** *The set of distinct north-step coarea values of P is $X(P)$. Consequently*

$$\boxed{\widetilde{\mathrm{run}}(P) = \min\{a \geq 1 : a\kappa \notin X(P)\}.} \tag{21}$$

Proof. Every north step in one vertical component has the same horizontal coordinate, and every component contains at least one north step. Thus the distinct coarea values are exactly the component columns. The second statement is the definition of ratio-run [2]. ☐

For $a \geq 1$, define

$$R_{\kappa,a} = \{\kappa, 2\kappa, \dots, (a-1)\kappa\}, \qquad F_{\kappa,a} = \{a\kappa\}.$$

Then $\widetilde{\mathrm{run}}(P) = a$ exactly when $R_{\kappa,a} \subseteq X(P)$ and $X(P) \cap F_{\kappa,a} = \varnothing$.

Theorem 6.2 (Composition-refined ratio-run with exact returns)**.** *For every $a \geq 1$ and $J \subseteq [d]$,*

$$\boxed{\#\{P \in \mathcal{D}_{m,\alpha} : \widetilde{\mathrm{run}}(P) = a,\ \mathrm{RetComp}(P) = J\} = \sum_{T \subseteq R_{\kappa,a}} (-1)^{|T|} \det M_J^{\alpha, F_{\kappa,a} \cup T}.} \tag{22}$$

Proof. The ratio-run condition is exactly the presence–absence condition above. Inclusion–exclusion over the required values, followed by Theorem 4.6, gives the formula. ☐

Corollary 6.3 (Composition-refined ratio-run–return enumeration)**.** *For $a, b \geq 1$,*

$$\boxed{\#\{P \in \mathcal{D}_{m,\alpha} : \widetilde{\mathrm{run}}(P) = a,\ \mathrm{ret}(P) = b\} = \sum_{\substack{J \subseteq [d] \\ |J| = b}} \sum_{T \subseteq R_{\kappa,a}} (-1)^{|T|} \det M_J^{\alpha, F_{\kappa,a} \cup T}.}$$

Example 6.4. For $(m,n) = (11,5)$ and $\alpha = (2,1,1,1)$, one has $\beta = (0,4,6,8)$ and $\kappa = 2$. Prescribe $\widetilde{\mathrm{run}} = 2$ and $J = \{1,4\}$. Then $R_{2,2} = \{2\}$ and $F_{2,2} = \{4\}$. The two inclusion–exclusion determinants are

$$\det M_{\{1,4\}}^{\alpha,\{4\}} = 6, \qquad \det M_{\{1,4\}}^{\alpha,\{2,4\}} = 3,$$

so the required number is $6 - 3 = 3$. The three component-column sequences are

$$(0,1,2,8), \qquad (0,2,3,8), \qquad (0,2,5,8).$$

7 Simultaneous run, ratio-run, and exact returns

Lemma 7.1 (Ordinary run as a first-missing component column)**.** *For every $P \in \mathcal{D}_{m,\alpha}$,*

$$\boxed{\mathrm{run}(P) = \min\{a \geq 1 : a \notin X(P)\}.}$$

Proof. By Lemma 6.1, $X(P)$ is the set of distinct coarea values. The ordinary run statistic is the first missing positive coarea value [2]. ☐

For positive integers a, c, define

$$R_{a,c} = \{1, \dots, a-1\} \cup \{\kappa, 2\kappa, \dots, (c-1)\kappa\}, \qquad F_{a,c} = \{a, c\kappa\}. \tag{23}$$

Theorem 7.2 (Simultaneous run–ratio-run–return-set enumeration)**.** *For $J \subseteq [d]$, if $R_{a,c} \cap F_{a,c} \neq \varnothing$, then*

$$\#\{P \in \mathcal{D}_{m,\alpha} : \mathrm{run}(P) = a,\ \widetilde{\mathrm{run}}(P) = c,\ \mathrm{RetComp}(P) = J\} = 0.$$

Otherwise,

$$\boxed{\#\{P \in \mathcal{D}_{m,\alpha} : \mathrm{run}(P) = a,\ \widetilde{\mathrm{run}}(P) = c,\ \mathrm{RetComp}(P) = J\} = \sum_{T \subseteq R_{a,c}} (-1)^{|T|} \det M_J^{\alpha, F_{a,c} \cup T}.} \tag{24}$$

Proof. The two first-missing conditions together are precisely $R_{a,c} \subseteq X(P)$ and $X(P) \cap F_{a,c} = \varnothing$. If the sets intersect, the requirements are inconsistent. Otherwise apply inclusion–exclusion and the exact-return determinant (15). □

Corollary 7.3 (Exact north-step return set)**.** *If $S \subseteq [n]$ is not of the form $S = \{A_{i-1}+1 : i \in J\}$ for some $J \subseteq [d]$, then no path of composition α has* ReturnSet$(P) = S$. *For such a set S and disjoint $R_{a,c}, F_{a,c}$, the number of paths with*

$$\mathrm{run}(P) = a, \qquad \widetilde{\mathrm{run}}(P) = c, \qquad \mathrm{ReturnSet}(P) = S$$

is the right-hand side of (24)*.*

Example 7.4 (Simultaneous run and ratio-run)**.** Again let $(m, n) = (11, 5)$ and $\alpha = (2, 1, 1, 1)$, so $\beta = (0, 4, 6, 8)$ and $\kappa = 2$. Prescribe run $= 3$ and $\widetilde{\mathrm{run}} = 2$. Then

$$R_{3,2} = \{1, 2\}, \qquad F_{3,2} = \{3, 4\}.$$

The admissible component-column sequences are

$$(0, 1, 2, 5),\ (0, 1, 2, 6),\ (0, 1, 2, 7),\ (0, 1, 2, 8).$$

The first three have RetComp $= \{1\}$, while the last has RetComp $= \{1, 4\}$. Hence the corresponding exact counts are 3 and 1.

Remark 7.5 (Automatic compatibility detection)**.** The condition $R_{a,c} \cap F_{a,c} = \varnothing$ automatically detects incompatibilities. For example, if $\kappa = 2$, then run $= 2$ requires column 2 to be absent while $\widetilde{\mathrm{run}} = 2$ requires it to be present, so the simultaneous class is empty.

8 Explicit enumeration at fixed ratio-signature

This section uses the first ratio-signature $\mathrm{sign}_1(P)$ of Dai–Fu–Qiu [2]. Their auxiliary path P_1 is formed by scanning P and collecting each κ-east-step block contributing to ratio-run or return. Thus P_1 is generally a shuffle-selected subpath rather than an initial segment. Let

$$\mathcal{D}_{m,\alpha,s} = \{P \in \mathcal{D}_{m,\alpha} : \mathrm{sign}_1(P) = s\},$$

and assume this class is nonempty. Write

$$r_{\alpha,s} = \widetilde{\mathrm{rr}}(\alpha, m, s),$$

for the minimum value attained by either $\widetilde{\mathrm{run}}$ or ret on this fixed-signature class in the notation of Dai–Fu–Qiu.

Remark 8.1 (The signature is not generally a prefix)**.** No argument below identifies a general P_1 with a prefix. The prefix interpretation arises only after passing, via the fixed-signature bijections of Dai–Fu–Qiu Theorem 3.7, to the representative with minimal return.

Lemma 8.2 (Minimal-return prefix collapse)**.** *Suppose $P \in \mathcal{D}_{m,\alpha,s}$ has $\widetilde{\mathrm{run}}(P) = a$ and $\mathrm{ret}(P) = b$. Set*

$$h = a + b - r_{\alpha,s}.$$

The fixed-signature bijections of Dai–Fu–Qiu reduce the class to a representative Q with

$$\widetilde{\mathrm{run}}(Q) = h, \qquad \mathrm{ret}(Q) = r_{\alpha,s}, \qquad \mathrm{sign}_1(Q) = s.$$

For this minimal-return representative, Q_1 is exactly the initial segment occupying the first $h\kappa$ east-step columns. Consequently

$$\boxed{|s| = h\kappa.} \tag{25}$$

Proof. Dai–Fu–Qiu Theorem 3.7 constructs fixed-signature bijections along the skew diagonal and states that the corresponding minimal value is $r_{\alpha,s}$ [2]. In the minimal-return representative, every return block already belongs to the initial family of blocks contributing simultaneously to ratio-run and return; there are no additional return-only blocks. The ratio-run value h contributes the first h sampled κ-blocks, so the union defining Q_1 is precisely the initial $h\kappa$ east-step segment. Since s is the ordinary east-step signature of Q_1, its length is $h\kappa$. □

Corollary 8.3 (Fixed-signature skew diagonal)**.** *If $\mathcal{D}_{m,\alpha,s} \neq \varnothing$, then $\kappa \mid |s|$. With*

$$h_s = \frac{|s|}{\kappa},$$

every nonempty fixed-signature statistic class satisfies

$$\boxed{\widetilde{\mathrm{run}}(P) + \mathrm{ret}(P) = h_s + r_{\alpha,s}.}$$

Let the positions of the 1's in s be

$$1 \le p_1 < \cdots < p_t \le |s|, \qquad t = |s|_1.$$

Lemma 8.4 (Prefix components determined by the signature)**.** *For a minimal-return representative Q as above, the first t component columns are forced:*

$$\boxed{x_i = p_i - 1 \qquad (1 \le i \le t).}$$

Proof. By Lemma 8.2, s is the ordinary east-step signature of the first $|s|$ east steps of Q. An east step has signature 1 exactly when it immediately follows a north step. The east step in signature position p_i therefore closes the ith vertical component, whose horizontal coordinate is $p_i - 1$. □

Corollary 8.5 (Reading the minimal return from the signature)**.** *For an admissible signature,*

$$\boxed{r_{\alpha,s} = \#\{i \in [t] : p_i - 1 = \beta_i\}.}$$

Proof. In the minimal-return representative, all returns occur among the forced prefix components. Apply Lemma 4.2 and Lemma 8.4. □

Lemma 8.6 (Forced separator and free suffix)**.** *For a minimal-return representative Q,*

$$x_{t+1} \ge h_s\kappa + 1 = |s| + 1$$

when $t < d$, and

$$x_{t+j} \le \beta_{t+j} - 1 \qquad (1 \le j \le d - t).$$

Proof. Because $\widetilde{\mathrm{run}}(Q) = h_s$, the sampled column $h_s\kappa$ is missing, so the next component after the prefix starts at least one column later. Since $\mathrm{ret}(Q) = r_{\alpha,s}$ and the prefix already accounts for all returns, no suffix component is a return; Lemma 4.2 gives the strict upper bound. $\square$

Put $p = d - t$ and define

$$c_j^{(\alpha,s)} = \beta_{t+j} - |s| - j - 1, \qquad 1 \le j \le p. \tag{26}$$

Then $c_1^{(\alpha,s)} \le \cdots \le c_p^{(\alpha,s)}$ because $\beta_{i+1} - \beta_i \ge 1$.

Theorem 8.7 (Explicit fixed-ratio-signature cardinality)**.** *Let s be admissible and use the notation above. Then*

$$\boxed{C_{m,\alpha,s} := |\{P \in \mathcal{D}_{m,\alpha,s} : \widetilde{\mathrm{run}}(P) = h_s,\ \mathrm{ret}(P) = r_{\alpha,s}\}| = \det_{1\le i,j\le d-t} \left[\binom{c_i^{(\alpha,s)}+1}{j-i+1}\right].} \tag{27}$$

The empty determinant is 1.

Proof. The signature fixes the prefix columns by Lemma 8.4. For the free suffix set

$$v_j = x_{t+j} - (|s| + j).$$

Lemma 8.6 gives

$$0 \le v_1 \le \cdots \le v_p, \qquad v_j \le c_j^{(\alpha,s)}.$$

The barrier is nondecreasing, so the classical bounded-sequence determinant gives (27). $\square$

Corollary 8.8 (Common cardinality on the fixed-signature skew diagonal)**.** *For every* $0 \le j \le h_s - r_{\alpha,s}$*,*

$$\boxed{|\{P \in \mathcal{D}_{m,\alpha,s} : \widetilde{\mathrm{run}}(P) = r_{\alpha,s} + j,\ \mathrm{ret}(P) = h_s - j\}| = C_{m,\alpha,s}.} \tag{28}$$

All other fixed-signature statistic classes are empty.

Proof. Apply the fixed-signature skew-diagonal bijection of Dai–Fu–Qiu Theorem 3.7 to the minimal-return class, then use Corollary 8.3 to exhaust the possible pairs. $\square$

Corollary 8.9 (Explicit fixed-signature polynomial)**.**

$$\boxed{\sum_{P\in\mathcal{D}_{m,\alpha,s}} u^{\widetilde{\mathrm{run}}(P)} v^{\mathrm{ret}(P)} = C_{m,\alpha,s}(uv)^{r_{\alpha,s}} h_{h_s - r_{\alpha,s}}(u,v).} \tag{29}$$

Example 8.10 (The fixed signature 11111110)**.** Let

$$(m,n) = (23,11), \qquad \alpha = (1,1,2,1,1,1,1,1,1,1), \qquad s = 11111110.$$

Then $\kappa = 2$, $|s| = 8$, $h_s = 4$, and $t = 7$. The component barrier is

$$\beta = (0,2,4,8,10,12,14,16,18,20),$$

and the signature forces $(x_1,\ldots,x_7) = (0,1,2,3,4,5,6)$. Hence $r_{\alpha,s} = 1$. The free suffix barrier is $(6,7,8)$, so

$$C_{23,\alpha,s} = \det\begin{pmatrix} 7 & 21 & 35 \\ 1 & 8 & 28 \\ 0 & 1 & 9 \end{pmatrix} = 154.$$

Therefore each of the four classes

$$(\widetilde{\mathrm{run}}, \mathrm{ret}) = (1,4), (2,3), (3,2), (4,1)$$

has cardinality 154, and the fixed-signature polynomial is $154\, uv\, h_3(u,v)$.

9 Paths below an arbitrary fixed boundary

Let B be a northeast lattice path whose north-step starting columns are

$$\mathbf{b} = (b_1, \ldots, b_n), \qquad b_1 \le \cdots \le b_n.$$

A path P lying weakly northwest of B has a north-step column sequence $u(P) \in \mathcal{A}_{\mathbf{b}}$. Thus Section 3 applies verbatim.

Corollary 9.1 (Arbitrary-boundary presence and exact contacts)**.** *Let $R, F \subset \mathbb{Z}_{>0}$ be finite and disjoint. For $J \subseteq [n]$,*

$$\boxed{\#\{P \le B : R \subseteq \operatorname{Val}(P),\ \operatorname{Val}(P) \cap F = \varnothing,\ C_B(P) = J\} = \sum_{T \subseteq R} (-1)^{|T|} \det \mathsf{M}_J^{\mathbf{b}, F \cup T}.}$$

9.1 Partial Dyck paths

Let $0 \le m \le n$ and let $\mathcal{B}_{m,n}$ be the paths from $(0,0)$ to (m,n) remaining weakly above $y = x$. Their north-step barrier is

$$\boxed{b_i = \min(i-1, m).}$$

For $i \le m+1$, contact $u_i = b_i = i-1$ is a diagonal contact; for $i > m+1$, contact $u_i = b_i = m$ is contact with the terminal vertical wall. The multivariate variables therefore separate two geometrically different kinds of boundary contact without changing the compression argument.

Define

$$\operatorname{Ret}_\Delta(P) = \{i \in [m+1] : u_i = i-1\}, \qquad \operatorname{ret}_\Delta(P) = |\operatorname{Ret}_\Delta(P)|.$$

Specializing $z_i = z$ for $i \le m+1$ and $z_i = 1$ afterward gives a polynomial $D^{\mathrm{par}}_{m,n;F}(z)$ satisfying

$$D^{\mathrm{par}}_{m,n;F}(z) = \sum_{\substack{P \in \mathcal{B}_{m,n} \\ \operatorname{Val}(P) \cap F = \varnothing}} z^{\operatorname{ret}_\Delta(P)}.$$

If

$$\operatorname{fm}(P) = \min\{a \ge 1 : a \notin \operatorname{Val}(P)\},$$

then $\operatorname{fm}(P) = a$ is the required/forbidden condition $R_a = \{1, \ldots, a-1\}$ and $F_a = \{a\}$. Hence the framework gives both the return-number polynomial and exact diagonal-return-set determinant sums for partial Dyck paths.

10 Column-increasing rational parking-function labelings

Classical rational parking functions are commonly represented as rational Dyck paths whose north steps carry labels increasing within each column; see Armstrong, Loehr, and Warrington [1]. For the present application, we use the same column-increasing labeling convention for the $m \times n$ paths considered here, without requiring coprimality. The path statistics run, $\widetilde{\operatorname{run}}$, and ret are inherited from the underlying path.

Lemma 10.1 (Labelings of a fixed composition)**.** *If $P \in \mathcal{D}_{m,\alpha}$, then the number of column-increasing labelings of its n north steps by $1, \ldots, n$ is*

$$\boxed{L(\alpha) = \frac{n!}{\alpha_1! \cdots \alpha_d!}.}$$

Proof. Choose which α_i labels occupy each vertical component. Once those sets are chosen, the increasing condition determines their within-column order uniquely. □

Define the fixed-composition labeled run–return polynomial

$$\mathcal{P}_{m,\alpha}(u,v) = \sum_{\pi\colon \mathrm{comp}(\underline{\pi})=\alpha} u^{\mathrm{run}(\pi)} v^{\mathrm{ret}(\pi)},$$

where $\underline{\pi}$ is the underlying path.

Theorem 10.2 (Composition-refined labeled run–return expansion)**.**

$$\boxed{\mathcal{P}_{m,\alpha}(u,v) = \frac{n!}{\alpha_1!\cdots\alpha_d!}(uv)^{r_\alpha} \sum_{K=r_\alpha}^{d} C_{\alpha,K} h_{K-r_\alpha}(u,v).} \tag{30}$$

Proof. Every underlying path of composition α has the same multiplicity $L(\alpha)$ by Lemma 10.1. Multiply Corollary 5.3 by this factor. ☐

Summing over all compositions gives an explicit labeled polynomial

$$\boxed{\mathcal{P}_{m,n}(u,v) = \sum_{\alpha\models n} \frac{n!}{\prod_i \alpha_i!}(uv)^{r_\alpha} \sum_{K=r_\alpha}^{\ell(\alpha)} C_{\alpha,K} h_{K-r_\alpha}(u,v).} \tag{31}$$

For ratio-run, let $\Sigma_{m,\alpha}$ be the admissible first ratio-signatures. Corollary 8.9 similarly gives

$$\boxed{\widetilde{\mathcal{P}}_{m,\alpha}(u,v) = \frac{n!}{\prod_i \alpha_i!} \sum_{s\in\Sigma_{m,\alpha}} C_{m,\alpha,s}(uv)^{r_{\alpha,s}} h_{h_s - r_{\alpha,s}}(u,v).} \tag{32}$$

Thus both labeled distributions have manifest positive complete-homogeneous expansions once the determinant coefficients are fixed.

Remark 10.3. No symmetric-function coefficient claim is needed here. The application uses only the standard column-increasing labeling rule and the fact that the statistics under consideration depend on the underlying path.

11 Computational validation

All results of the paper are established theoretically. As an independent check, we implemented exhaustive enumeration of the underlying bounded sequences and fixed-composition rational Dyck paths and compared the cardinalities with the determinant formulas.

For the general exact-contact theorem, every nondecreasing barrier of length at most 5 with entries in $\{0,1,2,3,4\}$ was tested, together with every forbidden subset of $\{1,2,3,4\}$ and every exact contact set. This produced 88,032 determinant–enumeration comparisons.

For the rational-path specializations, the ordinary fixed-composition formula was tested on all $m > n$ with $n \le 7$ and $m \le 12$, every composition of n, and every tested run–return class: 755 parameter instances and 9,983 statistic queries. The ratio-run formula with exact return components was tested on 628 parameter instances and 54,664 exact-return queries. The simultaneous run–ratio-run formula was tested on 501 parameter instances and 246,392 simultaneous queries. No discrepancy was found.

The fixed-ratio-signature construction was tested independently by forming sign_1 directly from the selected κ-column blocks in the Dai–Fu–Qiu definition, rather than by using the determinant reduction. The test covered 2,357 fixed-signature classes over 12,081 underlying paths, with no mismatch.

As a larger stress test, for

$$(m,n) = (23,11), \qquad \alpha = (1,1,2,1,1,1,1,1,1,1),$$

all 63,920 fixed-composition paths were enumerated. Exactly 616 have first ratio-signature 11111110, split into the four classes

$$(1,4),\ (2,3),\ (3,2),\ (4,1)$$

with exactly 154 paths in each, agreeing with Example 8.10. No mismatch occurred in any exhaustive test.

12 Conclusion

We developed a contact-preserving barrier-compression framework for refined lattice-path enumeration. The method separates value constraints, upper-bound geometry, and boundary contacts. Deleting forbidden values produces an order-preserving compression of the alphabet and the barrier, while a multivariate Lindström–Gessel–Viennot determinant retains the surviving contact positions. Inclusion–exclusion then incorporates arbitrary required values and simultaneous first-missing statistics.

The framework arose from a concrete arithmetic route. A completion-aware deterministic pushdown automaton for the 2:1 balance relation led to the scalar residual $\rho_{p,q} = qN_1 - pN_0$. For $m = gq$ and $n = gp$, the identity $R = g\rho_{p,q}$ converted the low-residual condition into the rational-path return condition, which in coarea coordinates became an upper-barrier contact. The present paper isolates the resulting compression mechanism from that discovery route and develops it as a self-contained enumerative method.

For rational Dyck paths with fixed ordered vertical-run composition, the method gives explicit formulas at several levels of refinement. Ordinary run–return classes reduce to one determinant. Ratio-run is handled by sampled presence–absence constraints and exact return-component determinants. Ordinary run and ratio-run can be prescribed simultaneously with exact return positions. At fixed ratio-signature, the minimal-return representative converts the signature data into a forced prefix and a bounded free suffix, giving a single determinant for the common cardinality on the associated skew diagonal. The same framework applies to arbitrary northeast boundaries and partial Dyck paths, while standard column-increasing labeling transfers the fixed-composition formulas to labeled rational parking-function enumerations.

All determinant identities are proved combinatorially, and independent exhaustive computation verified the general framework and the four rational-path specializations over broad finite parameter ranges. The discovery-to-enumeration route can therefore be summarized as

$$\boxed{\begin{array}{c}\text{pushdown residual} \longrightarrow \text{rank} \longrightarrow \text{boundary contact} \\ \longrightarrow \text{contact-preserving compression} \longrightarrow \text{explicit determinants}\end{array}}$$

Declarations

Funding. The author received no financial support for the research, authorship, or publication of this article.

Competing interests. The author declares that there are no competing interests.

Author contributions. The author is solely responsible for the conception of the work, development of the mathematical methods, proofs, computational validation, and preparation of the manuscript.

Data availability. No external datasets were used in this study. All mathematical results are derived within the manuscript.

Code availability. The computational validation was performed using a standalone exhaustive-enumeration program implementing the definitions and determinant formulas described in the manuscript. The validation code accompanies the manuscript as supplementary material.

Ethics approval and consent. Not applicable.

References

[1] D. Armstrong, N. A. Loehr, and G. S. Warrington, Rational parking functions and Catalan numbers, *Annals of Combinatorics* 20 (2016), 21–58. doi:10.1007/s00026-015-0293-6.

[2] L. Dai, S. Fu, and D. Qiu, Symmetric statistics on rational Dyck paths, *Advances in Applied Mathematics* 177 (2026), 103066. doi:10.1016/j.aam.2026.103066; arXiv:2503.20155.

[3] I. Gessel and G. Viennot, Binomial determinants, paths, and hook length formulae, *Advances in Mathematics* 58 (1985), no. 3, 300–321. doi:10.1016/0001-8708(85)90121-5.

[4] G. Kreweras, Sur une classe de problèmes de dénombrement liés au treillis des partitions des entiers, *Cahiers du Bureau universitaire de recherche opérationnelle, Série Recherche* 6 (1965), 9–107.

[5] R. Pemantle and H. S. Wilf, Counting nondecreasing integer sequences that lie below a barrier, *Electronic Journal of Combinatorics* 16 (2009), no. 1, Research Paper R60. doi:10.37236/149.